

Table I. Feature Implementation Methods by Model Type

Feature	RB	BF	DS	XGB	SMLP	TS	HB
Probabilistic Reasoning	Deterministic rules	Posterior calculations	Belief theory	Tree scoring → probabilities	Network outputs	Sequence logits	Neural + rule fusion
Historical Data	Flags / counters	Momentum tracking	Sequential mass combine	Lag features	Sliding window	Sequence history	History buffer
Safety Rules	Hard-coded	Threshold overrides	Signal overrides	Post safety checks	Rule fallback	Rule fallback	Safety override function
Training Strategy	Static Rules	Static Probabilistic Model	Static Evidential Model	Offline Retraining	Offline Training	Offline Training	Offline Training + Rule Override
Temporal Processing	Counters	Priors evolution	Sequential combine	Lag encoding	Short-history concat	1D convolutions	Buffer-based smoothing
Pretraining	—	Likelihood calibration	Mass calibration	Offline training	Synthetic data	Model initialization	Network pretrain
Gradient Learning	—	—	—	Gradient boosting	Back-propagation	Gradient descent	Backpropagation
Model Type	Rule-based	Probabilistic	Belief-based	Tree ensemble	Feedforward NN	Temporal NN	Hybrid NN + rules
Source(s): Authors' own creation							

Table II. Baseline Functional Performance Relative to the Expected Operating Point

Decision Model	Precision (95% CI)	Recall (95% CI)	F1-Score (95% CI)
Expected	0.7333	0.7333	0.7333
RB	0.7315 (0.7315, 0.7315)	0.7333 (0.7333, 0.7333)	0.7324 (0.7324, 0.7324)
BF	0.6861 (0.6861, 0.6861)	0.7000 (0.7000, 0.7000)	0.6895 (0.6895, 0.6895)
DS	0.7315 (0.7315, 0.7316)	0.7333 (0.7333, 0.7333)	0.7324 (0.7324, 0.7324)
XGB	0.7798 (0.7576, 0.8114)	0.8108 (0.7921, 0.8296)	0.7892 (0.7678, 0.8159)
SMLP	0.7784 (0.7592, 0.7976)	0.7908 (0.7675, 0.8108)	0.7824 (0.7595, 0.8016)
TS	0.6810 (0.6510, 0.7151)	0.6849 (0.6607, 0.7197)	0.6538 (0.6160, 0.6970)
HB	0.7653 (0.7489, 0.7785)	0.7653 (0.7486, 0.7785)	0.7653 (0.7486, 0.7785)
Source(s): Authors' own creation			

Table III. Scenario-Family F1-Score Relative to the Expected Operating Point

Decision Model	Parking (S1)	Gradual Approach (S2 - S4)	Sudden Obstacles (S5A - S5E)	Complex Motion (S6A - S6F)
Expected	0.7500	0.8333	0.5000	0.8750
RB	0.7491	0.8326	0.4992	0.8739
BF	0.0949	0.8326	0.4992	0.8739
DS	0.7491	0.8326	0.4992	0.8739
XGB	0.7437	0.8347	0.6363	0.8692
SMLP	0.8937	0.8428	0.6010	0.8954
TS	0.9952	0.7291	0.4868	0.7709
HB	0.7992	0.8485	0.6092	0.9158

Source(s): Authors' own creation

Table IV. Safety Reliability and Latency Profiles Relative to the Expected Operating Point

Decision Model	Mean Time-to-Detect (s)	Full_Brake FNR (95% CI)	Near-Miss Rate (NMR)
Expected	0.0000	0.0000	0.0001
RB	0.0510 (0.0506, 0.0516)	0.0000 (0.0000, 0.0000)	0.5278 (0.5276, 0.5281)
BF	0.0474 (0.0472, 0.0475)	0.0667 (0.0667, 0.0667)	0.4946 (0.4944, 0.4948)
DS	0.0509 (0.0506, 0.0513)	0.0000 (0.0000, 0.0000)	0.5278 (0.5277, 0.5280)
XGB	0.0210 (0.0101, 0.0346)	0.0000 (0.0000, 0.0000)	0.5362 (0.5305, 0.5440)
SMLP	0.7153 (0.4513, 0.9652)	0.0000 (0.0000, 0.0000)	0.5282 (0.5273, 0.5291)
TS	1.1694 (0.7566, 1.6890)	0.0070 (0.0002, 0.0203)	0.5273 (0.5255, 0.5292)
HB	0.2225 (0.0431, 0.4419)	0.0000 (0.0000, 0.0000)	0.5303 (0.5286, 0.5320)
Source(s): Authors' own creation			

Table V. Robustness Degradation under Simulated Sensor Noise

Decision Model	Noise Condition	Δ F1-Score		Δ FNR
		(Mean Drop)	Δ MTTD (s)	(Safety Penalty)
RB	Moderate	- 0.2663	- 0.0004	- 0.0000
	Severe	- 0.2661	- 0.0006	- 0.0000
BF	Moderate	- 0.2532	- 0.0039	+ 0.0068
	Severe	- 0.2373	- 0.0039	+ 0.0188
DS	Moderate	- 0.2626	+ 0.0025	- 0.0000
	Severe	- 0.2658	+ 0.0100	- 0.0000
XGB	Moderate	- 0.1007	+ 0.0662	- 0.0000
	Severe	- 0.0798	+ 0.0039	- 0.0000
SMLP	Moderate	- 0.1740	+ 0.3432	- 0.0000
	Severe	- 0.2057	+ 1.6772	- 0.0000
TS	Moderate	- 0.1355	+ 0.9781	- 0.2161
	Severe	- 0.1363	+ 1.1510	- 0.2975
HB	Moderate	- 0.1546	+ 0.0008	- 0.0000
	Severe	- 0.1792	+ 0.1327	- 0.0000
Source(s): Authors' own creation				